CARLO DAVID FURORE



Mechanical Engineering M.S. Candidate | Additive Manufacturing, Computational Mechanics and AI-Enabled Monitoring

Bologna / Milan, Italy | +39 344 179 9959 | U.S. and Italian dual citizen

carlodavid.furore@mail.polimi.it | alt: carlofurore@gmail.com | Native English and Italian

RESEARCH INTERESTS AND FIT

- Develop physics-informed monitoring algorithms that combine print-layer images with machine signals (force, pressure, vibration and process logs) to detect binder-distribution defects and quantify process drift in sinter-based additive manufacturing.
- Model the constitutive, rheology-informed and fracture response of green bodies, linking binder saturation, porosity, geometry and curing state to deformation or failure during depowdering, robotic handling and transfer before sintering.
- Long-term objective: close the loop between additive-process monitoring, inverse identification and design so that intended metamaterial behavior remains reliable despite manufacturing-induced porosity and anisotropy.

CURRENT ACADEMIC STATUS

M.S. Mechanical Engineering - Sustainable Manufacturing Engineering | Politecnico di Milano | *2025/26 first-year plan*

- **Progress:** 6 of 9 first-year courses completed; 40 of 60 ECTS earned; ECTS-weighted average 28.8/30 (30L counted as 30).
- Advanced Manufacturing Processes A - 30/30, 10 ECTS - Prof. Massimiliano Annoni.
- Advanced Materials for Mechanical Engineering - 30/30, 5 ECTS - Prof. Carlo Mapelli.
- De-Manufacturing - 30/30 cum laude, 5 ECTS - Prof. Marcello Colledani.
- Dynamics of Mechanical Systems - 26/30, 5 ECTS - Prof. Alberto Zasso.
- Measurements for Mechanical Engineering - 30/30, 5 ECTS - Prof. Emanuele Zappa.
- Methods for Sustainable Mechanical Design - 27/30, 10 ECTS - Prof. Mario Guagliano.
- **Remaining in the approved first-year plan (20 ECTS):** Design and Management of Production Systems - Prof. Sergio Terzi; Machine Design - Prof. Stefano Foletti; Manufacturing Systems Engineering II - Prof. Tullio Tolio.

Bachelor's Degree in Mechanical Engineering | Politecnico di Milano | *final grade 106/110*

- Thesis: Microplastic Filter Base and Flow-Management System - designed bypass/filter routing, counter-rotating gates, check valves and injection-molded/ultrasonically welded features for safe operation and maintainability.

SELECTED RESEARCH AND LABORATORY EXPERIENCE

Fracture Resistance of Cured Binder-Jetted Metal Parts | Laboratory of Mechanical Technology | *primary lab project*

- Under Prof. Paolo Parenti, designed and ran a test workflow for fragile binder-jetted specimens using custom geometry, CNC/G-code impact paths, a piezoelectric dynamometer and NI 9234 acquisition.
- Analyzed three-axis force signals at 10,240 Hz, isolated fracture peaks from motion artifacts and related pillar geometry/impact height to measured force variability.
- Interpreted failure as bending-dominated, estimated UTS near 2.3 MPa, and translated results into force envelopes for automated depowdering, handling and pre-sinter quality control.

Green/Sintered Materials Studies | binder jetting and multimaterial BMD | *supporting lab work*

- Contributed to porosity studies on green and sintered binder-jetted specimens and to sintering-property work on multimaterial Bound Metal Deposition solutions.

ENGINEERING SOFTWARE AND MONITORING TOOLS

BinderJetCalibrationAnalyzer | self-built Windows analysis application *| 2026*

- Built an image-analysis workflow for photographed calibration sheets that generates fill metrics, row/column profiles, build-surface risk maps, QC panels, JSON data and HTML reports.
- Converted visual operator checks into repeatable quantitative evidence, providing a practical base for future image/sensor fusion and anomaly-detection algorithms.

VoltScope Studio | NI accelerometer setup and signal-analysis application *| 2026*

- Created a guided workflow for configuring National Instruments accelerometer measurements, applying sensitivity corrections and generating repeatable monoaxial, triaxial and capacitive-channel time plots.

Injection Mold Estimator | cost and manufacturability application *| 2026*

- Developed an early-stage estimator linking geometry, material/process assumptions, cycle-time logic and mold considerations to rapid design-for-manufacturing decisions.

ADDITIONAL TECHNICAL PROJECTS

CNC Micromilling / Laser-Cutting Fit Study | manufacturing laboratory

- Generated G-code/toolpaths for a controlled geometry and compared a micromilled cavity with a laser-cut insert to study accuracy, tolerances and interference-fit feasibility.

Experimental Modal Analysis of a Canopy | dynamics assignment

- Identified four modes from measured FRFs using half-power bandwidth and MAC; the first mode agreed with FEM within about 2.4% in frequency (MAC 0.857).

FEM Analysis of a Bridge Model | dynamics assignment

- Implemented MATLAB beam FEM with mass/stiffness/Rayleigh damping matrices, modal truncation, harmonic response and moving-load integration with ode45.

TECHNICAL COMPETENCIES

Mechanics and modeling: Nonlinear/linear FEM, modal and vibration analysis, experimental signal interpretation, MATLAB, COMSOL Multiphysics, advanced SolidWorks FEM and Autodesk Fusion FEM.

Manufacturing: Binder jetting, BMD, green-part handling, sintering/porosity studies, CNC G-code, micromilling, laser cutting, Fusion CAM, injection molding and Fusion molding simulation.

Software and AI: Python, MATLAB, C/C++, JavaScript, HTML/CSS, Dart/Flutter, Electron, image processing, CSV/signal analysis, Codex/OpenAI-assisted development and AI-native rapid prototyping.

Materials and sustainability: Granta materials selection, SimaPro LCA, Thermo-Calc fundamentals, Minitab and DFM/DFA.

Measurement systems: National Instruments hardware workflows, piezoelectric force measurement, accelerometers and repeatable data-acquisition/reporting pipelines.

INTERESTS AND ADDITIONAL STRENGTHS

- Competitive chess: FIDE-rated player; sustained practice in planning, pattern recognition and decision-making under time pressure.
- Conservatory-trained pianist: 10 years of formal study, including 7 years at a conservatory, with advanced training in technique, interpretation, repertoire and performance.
- Independent software creation: build packaged applications from problem definition to executable release, including manufacturing tools and earlier game-development utilities.
- U.S. and Italian dual citizen; native English and Italian, basic French; work-authorized in both the United States and European Union.